

Literature Summary

The initial deliverable is a Literature Summary addressing various subtopics: probing methodology and the insights probing classifiers provide about representations; the hierarchical organization in embeddings and the understanding of how abstract and concrete concepts are represented; weak supervision techniques and how document structure may serve as an alternative to manual annotation; and ultimately the relevance to SLoD, which includes the insights that most effectively aid in identifying abstraction levels in scientific writing.

The works on probing and representation analysis, including Tenney, Das, and Pavlick's "BERT Rediscovered the Classical NLP Pipeline" and Belinkov's "Probing Classifiers: Promises, Shortcomings, and Advances," along with those focused on hierarchical and hyperbolic representations like Nickel and Kiela's "Poincaré Embeddings for training Hierarchical Representations" and Vilnis and McCallum's "Word Representations via Gaussian Embedding," as well as research on scientific text and weak supervision such as Lo et al.'s "S2ORC: The Semantic Scholar Open Research Corpus" and Ratner et al.'s "Snorkel: Rapid Training Data Creation with Weak Supervision," collectively provide a foundation for comprehending probing classifiers as diagnostic tools. These tools may be utilized to determine if a certain characteristic can be predicted from a model's internal representations of the input. This is beneficial as it provides us a method to understand large language models and pretrained encoders. These papers likewise clarify how this instrument can be utilized, the ways it can be implemented, and how the outcomes ought to be interpreted. The initial significant subject is probing methodology. A probing classifier is employed when there is a pretrained model, and we aim to determine if it holds information regarding a specific feature of the input. Researchers have employed this approach to examine if models represent various parts of speech, morphological features, syntactic attributes, semantic characteristics, and other linguistic details. This project aims to investigate if we can identify the level of abstraction in scientific texts using frozen embeddings.

The mechanism functions in this manner. We begin with a specific input, such as a portion of text from a research paper, aiming to categorize it to determine if certain information can be extracted from the model's representation. The input is processed by an encoder, or pretrained model, generating frozen embeddings. These embeddings serve as representations of the provided text. A distinct classifier is trained based on these representations. This classifier takes the embeddings as input and forecasts the category related to the property we wish to examine.

This classifier should not be overly complex since a more complex classifier may reach greater accuracy, yet it might also memorize text patterns rather than indicating if the target property is genuinely decodable from the representation. Consequently, probing studies

typically employ basic classifiers like logistic regression or linear SVM. In this project, the focus is on the semantic level of detail; specifically, the aim is to assess whether the level of abstraction in a text can be identified by categorizing a specific text segment into macro, meso, or micro levels. A well-performing probing classifier indicates that the target property can be extracted from the representation. The most precise interpretation is that the property can be derived from the embedding space. Nonetheless, this does not imply that the model truly comprehends this characteristic or utilizes it in producing output.

Interpreting probing results also carries risks. Elevated performance might be related to other characteristics of the text instead of semantic abstraction alone; hence, the literature highlights the necessity of supplementary control analysis. Probing outcomes must be contrasted with baselines; for instance, a majority-class baseline indicates the performance possible without effectively utilizing embeddings. The second topic concerns hierarchical structures in embeddings, with research indicating that representations can capture hierarchy, asymmetrical relations, specificity, uncertainty, and various complex structures. This is significant since SLoD is inherently structured at various levels of abstraction. On a macro scale, text can include overarching assertions, research advancements, final thoughts, and overall inspiration. As the text grows more detailed, it incorporates additional specific information, including numbers, methodological processes, model specifics, experimental conditions, metrics, and technical outcomes.

To conduct the experiment, it's essential to have a data set for training and evaluating the classifier, which is why we present the idea of weak supervision. It involves creating training labels through rules or signals rather than through manual annotation. Manual tagging can be costly and require significant time as it depends on human effort. Ratner et al. developed Snorkel as a framework to generate labeled data via labeling functions, which are programmatic rules that designate labels to data points. The guidelines are not ideal since research articles do not consistently employ the same table of contents or section titles, making it challenging to depend solely on programmed keywords or section names for data labeling. Within a weak supervision framework, we can apply criteria like section titles, the organization of the table of contents, frequently occurring words in abstracts or detailed papers, along with technical or numerical indicators conveyed through methods, experiments, and comprehensive outcomes.

For instance, the macro label can be designated to sections such as title, abstract, introduction, and conclusion. Lead paragraphs in sections can receive the meso label as they frequently present the aim or subject matter of that section. Methods, experiments, and paragraphs with detailed results can be given the micro label as they typically include operational, technical, and numerical specifics. Thus, rules based on sections may be utilized to acquire data for training.

The Semantic Scholar Open Research Corpus offers an extensive collection of scientific papers that include metadata, abstracts, references, and organized full text. By maintaining the structure of scientific papers, it assists in automatically extracting text segments and labeling them according to their location within the paper.

The SLoD task integrates three research concepts: probing classifiers, hierarchical structures, and weak supervision derived from structured scientific text. Probing methodology offers the experimental framework and demonstrates how to create the experiment. Weak supervision provides the labeling approach and assists in acquiring training and test datasets. Research on hierarchical representation provides the conceptual foundation for considering abstraction levels as aspects that can be represented in embeddings.

In this configuration, the encoder stays unchanged. It does not train on the SLoD task; it solely generates embeddings. The probing classifier undergoes training using the training embeddings and is subsequently evaluated on new test embeddings. The most significant outcome would be excellent performance across three scenarios: in-domain, cross-domain, and length-controlled. In-domain performance indicates if the signal can be retrieved within a specific scientific domain. Cross-domain performance evaluates if the signal extends beyond vocabulary specific to a domain. Length-controlled performance evaluations determine if the probe utilizes semantic abstraction instead of depending on non-semantic attributes like text length.

The primary constraints must also be considered. Sections such as abstracts, introductions, and methods in scientific papers can vary in length. If length is unmanaged, the classifier could pick up on patterns tied to length. Vocabulary specific to a section presents an additional risk. Macro sections can include expressions like “we suggest,” “we present,” “our contribution,” or “the aim of this study.” Micro sections could include terms like “table,” “accuracy,” “dropout,” “training,” “test,” and numerical data. A further risk is domain leakage, as training and testing the model solely on one domain may lead it to adopt writing patterns specific to that domain. Cross-domain evaluation assesses if the acquired signal is transferable to a different area.

It is crucial that scientific papers utilized for training do not become part of the test dataset. Text segments from the same document must not be present in both training and testing datasets. Alternatively, the classifier might pick up the vocabulary, topic, or writing style of a specific paper instead of a general SLoD signal. Consequently, the dataset must be divided by paper ID instead of being randomly segmented by text span. In general, the literature indicates that probing classifiers may help ascertain if SLoD can be decoded from frozen embeddings, though the findings need careful interpretation and consideration of potential confounds.